

Derivation of the Exponential Moving Average

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1 Overview

Picture a stream of data arriving one value at a time – the daily price of a stock, an hourly temperature reading. Two things are wanted of a running summary of such a stream. It should track the *recent* behavior of the series, discounting old data smoothly rather than abruptly. And it should be cheap to update: when the next value arrives, we should not have to recompute a large sum from scratch. The exponential moving average delivers both, and this paper derives it – together with a matching local standard deviation and the correction factor that makes the variance estimate unbiased.

The essence of the *exponential moving average* (EMA) is to provide a *localized* estimate of a time series as it proceeds. It does this by using a weighted average of the values in a “window” around each point of the series in a way that each summand in the average is multiplied by powers of a factor, λ , in the interval $(0, 1)$. In what follows we will assume a 1-based index, and that the data series $\mathbf{x}$ has length N .

Specifically, the exponential moving average (over a window of length m) is defined as:

$$e_n = \sum_{k=0}^{m-1} x_{n-k} w_{k+1} \tag{1}$$

To get the *exponential decay* in weights we assume that $w_{k+1} = \lambda w_k$.¹ Since the parameter λ is in the interval $(0, 1)$, it represents a decay factor. Notice that the indices of $\mathbf{x}$ and that of $\mathbf{w}$ go in the “opposite direction”. This is done so that the largest weights are associated with the most recent data in the averaging window.

Before we go further, there is a problem with the formula as stated. If $n = 1$; or, in fact, any value less than m , the expression x_{n-k} does not make sense for $k \geq n$. (At $n = m$ the window is exactly full: its oldest term is x_1 .) That is, at the start of the computation and for a little while, there is not a full window of values over a window of length m . As there is no information before x_1 , we interpret these values to be x_1 . We will apply this principle more generally below. That is, if $\mathbf{z}$ is a data series, we extend the definition of $\mathbf{z}$ so that it is defined for indices less than 1 as:

¹The reason this is called exponential is that it is the discrete analog of continuous exponential decay. To see this let $f(t) = e^{-\beta t}$ for $\beta > 0$. Then sampling this function at regular intervals of say 1 we get the sequence: $\{e^{-\beta}, e^{-2\beta}, e^{-3\beta}, \dots\}$. Here we can see that the elements of the sequence are powers of a value, $\lambda = e^{-\beta}$. That is, the sequence is nothing other than: $\{\lambda, \lambda^2, \lambda^3, \dots\}$.

$$z_k \equiv z_{\max(k, 1)} \quad (2)$$

While we can use (1) to compute the EMA it is *inefficient*: each new point costs m multiplications, so the full series costs on the order of Nm operations – and the cost of keeping the average local grows with how local we want it to be. The purpose of this paper is to derive a recursive formula that is fast to compute: a fixed handful of operations per point, *independent* of the window length m . The formula takes advantage of the fact that when the weights are exponentials (powers of λ) we don't have to recompute the bulk of the sums in the averaging window.

Regardless of the averaging scheme, we want our weights to sum to one.² But we also want the weights to be exponential; that is, $w_{k+1} = \lambda w_k$.

Ideally, all we want to do is give the decay factor, λ . Given λ , the weights are constructed as follows.

For $i \in [1, m]$ define

$$\hat{w}_i = \lambda^i \quad (3)$$

The $\hat{w}$ have the property that they decay with λ ; that is, $\lambda \hat{w}_k = \hat{w}_{k+1}$. However, how do we ensure that these values sum to 1 over the window?

To do this, we simply normalize the weights:

$$w_i = \frac{\hat{w}_i}{\sum_{k=1}^m \hat{w}_k} \quad \forall i \in [1, m] \quad (4)$$

Consequently,

$$\sum_{i=1}^m w_i = 1 \quad (5)$$

In doing this, did the proposed weights, $\mathbf{w}$, lose the decay factor property? The answer is no, the property remains:

$$\lambda w_i = w_{i+1} \quad \forall i \in [1, m-1]. \quad (6)$$

More generally, we can find *all* weights that have a decay factor of λ as they are solutions of the following difference equation:

$$w_{n+1} = \lambda w_n \quad (7)$$

$$w_1 = \alpha \quad (8)$$

²If we consider the underlying data to be generated from a series of identically distributed random variables; then, treating the series elements as random variables we can view the moving average as an estimate of their mean, μ . To be a good estimate we would like it to be *unbiased*: $\mathbb{E} \left[\sum_{k=0}^{m-1} x_{n-k} w_{k+1} \right] = \mu$. By linearity of the expectation, $\mathbb{E} \left[\sum_{k=0}^{m-1} x_{n-k} w_{k+1} \right] = \sum_{k=0}^{m-1} \mathbb{E} [x_{n-k}] w_{k+1} = \mu \sum_{k=1}^m w_k$; so the estimate is unbiased exactly when the weights sum to 1.

This is the geometric difference equation – the discrete analog of exponential decay – and its solution is:

$$w_n = \alpha \lambda^{n-1} \quad n \geq 1 \quad (9)$$

However, we need the weights to be *normalized* over a window of length, m :

$$\sum_{i=1}^m w_i = 1 \quad (10)$$

Plugging in our general formula for an exponential sequence we have:

$$\sum_{i=1}^m \alpha \lambda^{i-1} = 1 \quad (11)$$

$$\alpha \frac{1 - \lambda^m}{1 - \lambda} = 1 \quad (12)$$

Therefore, a decay factor, λ , and averaging window length, m , determine *the* initial weight, α , and thus determine *the* associated normalized weights:

$$\alpha = \frac{1 - \lambda}{1 - \lambda^m} \quad (13)$$

Note. We will use (2), (5), (6), and (13) in our derivations below.

2 Recursive Formula for EMA

We state again the definition of the exponential moving average of x as:

$$e_n = \sum_{k=0}^{m-1} x_{n-k} w_{k+1} \quad (14)$$

In order to find a recursive formula, we examine the next element in the smoothing sequence and try to relate it to the present element at index n .

$$\begin{aligned}
e_{n+1} &= \sum_{k=0}^{m-1} x_{n+1-k} w_{k+1} \\
&= \sum_{k=0}^{m-1} x_{n-(k-1)} w_{k+1} \\
&= \sum_{k=-1}^{m-2} x_{n-k} w_{k+2} \\
&= \sum_{k=0}^{m-2} x_{n-k} w_{k+2} + x_{n+1} w_1 \\
&= \lambda \sum_{k=0}^{m-2} x_{n-k} w_{k+1} + x_{n+1} w_1 \quad (\text{Since } \lambda w_{k+1} = w_{k+2}) \\
&= \lambda \sum_{k=0}^{m-1} x_{n-k} w_{k+1} + x_{n+1} w_1 - \lambda x_{n-(m-1)} w_m \\
&= \lambda \sum_{k=0}^{m-1} x_{n-k} w_{k+1} + x_{n+1} w_1 - \lambda x_{(n+1)-m} w_m
\end{aligned}$$

Finally, we may write³

$$e_{n+1} = \lambda (e_n - x_{(n+1)-m} w_m) + \alpha x_{n+1} \quad \forall n \in [1, N-1] \quad (15)$$

$$e_1 = x_1 \quad (16)$$

In words: as the window slides one step, three things happen. The old average is decayed by λ – which promotes every weight one slot, turning each w_{k+1} into w_{k+2} ; the datum that slid out the back of the window is removed (the $x_{(n+1)-m} w_m$ term); and the new datum is admitted at the top weight, α . One could have guessed a formula of this shape from the picture of a sliding window alone; the algebra above confirms the guess and pins down the coefficients.

It is worth naming the game being played here. Writing $b_n = \alpha x_{n+1} - \lambda w_m x_{(n+1)-m}$, the result reads $e_{n+1} = \lambda e_n + b_n$ – a first-order linear difference equation, the same equation $x_{n+1} = \alpha x_n + b_n$ that models, say, a loan being paid down, and whose closed-form solution is found by summing factors. The usual traffic on that road runs from recursion to closed form; here we have traveled it in reverse, from the closed form (a windowed sum) back to the recursion. In both directions the trip is powered by the same fact: a geometric sequence is self-similar under a shift of index.

As a calibration, feed the recursion a series it cannot mistake: the constant series $x_n = c$. The definition (1) gives $e_n = c$, since the weights sum to one. The recursion agrees: $\lambda(c - c w_m) + \alpha c = c(\lambda - \alpha \lambda^m + \alpha) = c$, using $w_m = \alpha \lambda^{m-1}$ and $\alpha(1 - \lambda^m) = 1 - \lambda$. The new machine reproduces the answer we already own – which is how one learns to trust a new tool.

³From the definition of e_n and using (2), (5), and (13), $e_1 = \sum_{k=0}^{m-1} x_{1-k} w_{k+1} = x_1 \sum_{k=0}^{m-1} w_{k+1} = x_1 \sum_{k=1}^m w_k = x_1$.

The corresponding 0-based index formula is:⁴

$$e_n = \lambda(e_{n-1} - x_{n-m}w_{m-1}) + \alpha x_n \quad \forall n \in [1, N-1] \quad (17)$$

$$e_0 = x_0 \quad (18)$$

3 Recursive Formula for a Local Standard Deviation

In practice, one also wants to know a localized standard deviation of the process. In applications, one can provide “Bollinger” bands around the EMA estimate. These bands are (typically) some multiple of the standard deviation of the process.

We derive a localized standard deviation of the process starting with the definition of the exponential moving average and its moving variance.⁵

$$e_n = \sum_{k=0}^{m-1} x_{n-k} w_{k+1} \quad (19)$$

$$v_n = \sum_{k=0}^{m-1} (x_{n-k} - e_{n-k})^2 w_{k+1} \quad (20)$$

Note that the variance (20) measures squared deviation of each x_{n-k} from its own time’s EMA e_{n-k} , rather than from the current e_n . This choice lets the recursive update for v_n be derived using the same machinery as e_n .⁶

Note, that the form of the two equations is identical; that is, they both have the form:

$$g_n = \sum_{k=0}^{m-1} f(x_{n-k} - c_{n-k}) w_{k+1} \quad (21)$$

In equation (19), $c_{n-k} \equiv 0$; while in equation (20), $c_{n-k} = e_{n-k}$.

One can find a recursive formula g_n in *exactly* the same way as was done for (19). This is done by reworking the derivation of the recursive formula for (19), replacing x_{n-k} with $f(x_{n-k} - c_{n-k})$.

The recursive formula for g_n is

$$g_{n+1} = \lambda(g_n - f(x_{(n+1)-m} - c_{(n+1)-m})w_m) + \alpha f(x_{n+1} - c_{n+1}) \quad \forall n \in [1, N-1] \quad (22)$$

$$g_1 = f(x_1 - c_1) \quad (23)$$

⁴We write the formulas down for both 1- and 0-based vector indices because computer languages typically use one or the other. In the 0-based formulas the weights are re-indexed as well: they are $w_0, \dots, w_{m-1}$, so the first (largest) weight is $\alpha = w_0$ and the oldest weight is w_{m-1} .

⁵This variance formula needs to be corrected by a factor which we leave to the end. The correction is a factor that is used to make the estimated standard deviation of the moving average *unbiased*. The formula for this factor is independent of the structure of the weights. This means that it can be pre-computed independently of the data series.

⁶The unbiasing factor $1 - \sum_{i=1}^m w_i^2$ derived in the appendix assumes a single common center, $\bar{x} = \sum_i x_i w_i$, applied to every summand. Here each summand uses its own center e_{n-k} , so the correction is applied as a working approximation; an exact unbiased correction for centers e_{n-k} would require a separate derivation.

The 0-based index formula is:

$$g_n = \lambda (g_{n-1} - f(x_{n-m} - c_{n-m})w_{m-1}) + \alpha f(x_n - c_n) \quad \forall n \in [1, N-1] \quad (24)$$

$$g_0 = f(x_0 - c_0) \quad (25)$$

The corresponding f for (20) is $f(x) = x^2$; consequently, its recursive formula is

$$v_{n+1} = \lambda (v_n - (x_{(n+1)-m} - e_{(n+1)-m})^2 w_m) + \alpha (x_{n+1} - e_{n+1})^2 \quad \forall n \in [1, N-1] \quad (26)$$

$$v_1 = (x_1 - e_1)^2 = 0 \quad (27)$$

The corresponding 0-based index formula is:

$$v_n = \lambda (v_{n-1} - (x_{n-m} - e_{n-m})^2 w_{m-1}) + \alpha (x_n - e_n)^2 \quad \forall n \in [1, N-1] \quad (28)$$

$$v_0 = (x_0 - e_0)^2 = 0 \quad (29)$$

The dictionary in one place – one recursion, several customers:

quantity	$f(x)$	center c_{n-k}	initial value
mean e_n	x	0	$e_1 = x_1$
variance v_n	x^2	e_{n-k}	$v_1 = 0$
general g_n	any f	any series	$g_1 = f(x_1 - c_1)$

Any localized statistic of the form (21) rides the same machine; for instance, a localized third moment takes $f(x) = x^3$ with center e_{n-k} , and the appendix prices the bias correction for such higher moments as well.

Finally, the localized estimate of the standard deviation of the *process* is:⁷

$$s_n = \sqrt{\frac{v_n}{1 - \sum_{i=1}^m w_i^2}} \quad (30)$$

The variance corrective factor⁸, $1 - \sum_{i=1}^m w_i^2$, is a one-time calculation independent of n . This factor is the standard correction when determining empirical variance of a weighted average.

Note. In practice, one may wish to use some prior estimate of the standard deviation of the process and smoothly move from that to the estimate above over the averaging windows when $n \in [1, m]$. For the same reason – $v_1 = 0$ and a partially filled window – the early values s_n , for n up to about m , should not be relied upon on their own.

⁷The window length, m , must be greater than 1. Note that this is an estimate of the local standard deviation of the underlying process – which is what Bollinger bands require – not the standard deviation of the estimator e_n itself; the latter is the smaller quantity $\sigma \sqrt{\sum_{i=1}^m w_i^2}$.

⁸This factor is derived in the appendix.

4 How to Specify the λ Factor

When using the EMA, practitioners do not specify the factor λ directly; instead they provide the “half-life”, H , of the decay. The *half-life* is the index H such that $w_H/w_1 = 1/2$; that is, the weight halves in going from index 1 to index H .⁹ Substituting $w_n = \alpha\lambda^{n-1}$ gives:

$$\frac{w_H}{w_1} = \frac{\alpha\lambda^{H-1}}{\alpha} = \lambda^{H-1} = \frac{1}{2} \quad (31)$$

For a given λ , an integer H satisfying this exactly may not exist; in that direction one takes the closest integer.¹⁰

In our case we are interested in the reverse; namely, given a half-life, H , determine the decay factor λ . To do this we solve the equation $\lambda^{H-1} = \frac{1}{2}$ for λ . Taking the log of this equation we have:¹¹

$$\begin{aligned} \log(\lambda^{H-1}) &= \log\left(\frac{1}{2}\right) \\ (H-1)\log(\lambda) &= \log(1) - \log(2) \\ \log(\lambda) &= -\frac{\log(2)}{H-1} \end{aligned}$$

The formula to determine λ given the half-life H is:

$$\lambda = e^{-\frac{\log(2)}{H-1}} \quad (32)$$

For example, if the window of the moving average is 10, you might decide that the half-life occurs around the middle of the window. Setting H to be 5, the decay factor, λ is $e^{-\frac{\log(2)}{4}} \approx 0.8409$.

⁹This convention measures the halving over $H-1$ steps. Some libraries (e.g., the `ewm` functions of pandas) instead define the half-life by $\lambda^H = 1/2$ – halving over H steps – which yields $\lambda = 2^{-1/H}$. The two conventions differ by one step: for $H = 5$, ours gives $\lambda \approx 0.8409$ while the other gives $\lambda = 2^{-1/5} \approx 0.8706$.

¹⁰Note that H must be greater than 1 and no more than the window length, m .

¹¹The log function used below is the logarithm base e .

A Unbiased Estimator for a Sequence of IID Random Variables

In this section we derive the corrective formula for computing the empirical variance.

In what follows we assume that we are given a sequence of independent identically distributed random variables (IID): $\{x_i\}_{i=1}^N$ and corresponding weights: $\{w_i\}_{i=1}^N$. We assume that the random variables have a mean and a variance:

$$\mathbb{E}[x_i] = \mu \quad \forall i \in [1, N] \quad (33)$$

$$\mathbb{E}[(x_i - \mu)^2] = \sigma^2 \quad \forall i \in [1, N] \quad (34)$$

We assume that the weights are non-negative and are normalized:¹²

- $\sum_{i=1}^N w_i = 1$
- $\forall i, 0 \leq w_i < 1$

If we take the expectation of the weighted sum, $\sum_{i=1}^N x_i w_i$ we find that it is μ .¹³

$$\mathbb{E}\left[\sum_{i=1}^N x_i w_i\right] = \sum_{i=1}^N \mathbb{E}[x_i w_i] = \sum_{i=1}^N \mathbb{E}[x_i] w_i = \mu \sum_{i=1}^N w_i = \mu \quad (35)$$

The weighted sum represents an estimate of the true mean of the IID random variables, which is μ . And in this case we see that it is a good estimate in that the expected value of the weighted combination of the random variables is μ .

We refer to such “good” estimates as *unbiased estimators*.

Note what this computation shows: *every* choice of normalized weights gives an unbiased estimate of the mean, so unbiasedness leaves the weights free to serve another master. The companion paper “Properties of a Linear Unbiased Minimum Variance Estimator” chooses them to minimize the variance of the estimate – for measurements of unequal precision the answer there is inverse-variance weighting, $w_i \propto 1/\sigma_i^2$. The exponential weights of the present paper are chosen instead to localize the average in time: they favor recency, not precision. Locality is bought with variance.

Now consider the sum $\sum_{i=1}^N (x_i - \mu)^2 w_i$. People use this as an estimate of the variance of the random variables. Is this a “good” (unbiased) estimate? Yes, it is as: $\mathbb{E}\left[\sum_{i=1}^N (x_i - \mu)^2 w_i\right] = \sum_{i=1}^N \mathbb{E}[(x_i - \mu)^2] w_i = \sigma^2 \sum_{i=1}^N w_i = \sigma^2$.

But what if we replace the true mean, μ , above with our estimate: $\bar{x} = \sum_{i=1}^N x_i w_i$? In this case it turns out that the resulting estimate for the variance is *biased*. However, it happens that we are “off” by a certain factor and that factor depends only on the weights we use.¹⁴

¹²We also assume that the weight is not concentrated at a single index.

¹³This is due to the linearity of the expectation operator: $\mathbb{E}[ax + by + c] = a\mathbb{E}[x] + b\mathbb{E}[y] + c$, provided a, b, c are constants.

¹⁴In fact, the factor doesn’t depend on the ordering of the weights.

To find this factor we examine the expected value of the straightforward formula one would imagine using to estimate the variance of the random variables.

$$\begin{aligned}
\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^2 w_i \right] &= \sum_{i=1}^N \mathbb{E} [(x_i - \bar{x})^2] w_i \\
&= \sum_{i=1}^N \mathbb{E} [(x_i - \mu) + (\mu - \bar{x})]^2 w_i \\
&= \sum_{i=1}^N \mathbb{E} [(x_i - \mu)^2 - 2(x_i - \mu)(\bar{x} - \mu) + (\bar{x} - \mu)^2] w_i \\
&= \sum_{i=1}^N (\mathbb{E} [(x_i - \mu)^2] + \mathbb{E} [(\bar{x} - \mu)^2] - 2\mathbb{E} [(x_i - \mu)(\bar{x} - \mu)]) w_i
\end{aligned} \tag{36}$$

We separately compute the expectation of each of the three summands.

$$\mathbb{E} [(x_i - \mu)^2] = \sigma^2 \tag{37}$$

$$\begin{aligned}
\mathbb{E} [(\bar{x} - \mu)^2] &= \mathbb{E} \left[\left(\left(\sum_{j=1}^N x_j w_j \right) - \mu \right)^2 \right] \\
&= \mathbb{E} \left[\left(\sum_{j=1}^N (x_j w_j - \mu w_j) \right)^2 \right] \\
&= \mathbb{E} \left[\left(\sum_{j=1}^N (x_j - \mu) w_j \right)^2 \right] \\
&= \mathbb{E} \left[\left(\sum_{j=1}^N (x_j - \mu) w_j \right) \left(\sum_{k=1}^N (x_k - \mu) w_k \right) \right] \\
&= \mathbb{E} \left[\sum_{j=1}^N \sum_{k=1}^N (x_j - \mu)(x_k - \mu) w_j w_k \right] \\
&= \left(\sum_{j=1}^N \sum_{k=1}^N \mathbb{E} [(x_j - \mu)(x_k - \mu)] w_j w_k \right) \\
&= \left(\sum_{j=1}^N \sum_{k=1}^N \sigma^2 w_j w_k \delta_{j,k} \right) \\
&= \left(\sum_{j=1}^N \sigma^2 w_j^2 \right) \\
&= \sigma^2 \left(\sum_{j=1}^N w_j^2 \right)
\end{aligned} \tag{38}$$

$$\begin{aligned}
\mathbb{E}[(x_i - \mu)(\bar{x} - \mu)] &= \mathbb{E}\left[(x_i - \mu)\left(\left(\sum_{j=1}^N x_j w_j\right) - \mu\right)\right] \\
&= \mathbb{E}\left[(x_i - \mu)\left(\sum_{j=1}^N (x_j - \mu)w_j\right)\right] \\
&= \mathbb{E}\left[\sum_{j=1}^N (x_i - \mu)(x_j - \mu)w_j\right] \\
&= \sum_{j=1}^N \mathbb{E}[(x_i - \mu)(x_j - \mu)] w_j \\
&= \sum_{j=1}^N \sigma^2 w_j \delta_{i,j} \\
&= \sigma^2 w_i
\end{aligned} \tag{39}$$

Substituting these three results into (37):

$$\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^N (x_i - \bar{x})^2 w_i\right] &= \sum_{i=1}^N \left(\sigma^2 + \sigma^2 \sum_{j=1}^N w_j^2 - 2\sigma^2 w_i\right) w_i \\
&= \sigma^2 + \sigma^2 \sum_{j=1}^N w_j^2 - 2\sigma^2 \sum_{i=1}^N w_i^2 \\
&= \sigma^2 \left(1 - \sum_{i=1}^N w_i^2\right)
\end{aligned}$$

Therefore, to make $\sum_{i=1}^N (x_i - \bar{x})^2 w_i$ an unbiased estimator of the variance σ^2 we need to correct by dividing by the factor $\left(1 - \sum_{i=1}^N w_i^2\right)$.¹⁵

Consequently, $\frac{1}{1 - \sum_{i=1}^N w_i^2} \sum_{i=1}^N (x_i - \bar{x})^2 w_i$ is an unbiased estimator for the variance of any of the IID variables: $\{x_i\}_{i=1}^N$. The unbiased standard deviation is then

$$\sqrt{\frac{1}{1 - \sum_{i=1}^N w_i^2} \sum_{i=1}^N (x_i - \bar{x})^2 w_i} \tag{40}$$

¹⁵Note, this factor is non-zero when the weights are not concentrated at a single index. As a consequence, the formula requires $N > 1$.

What is the formula when the weights are evenly distributed? That would mean that the weights are $w_i = \frac{1}{N}$. In this case the estimate for the standard deviation of the random variables is:

$$\begin{aligned}
\sqrt{\frac{1}{1 - \sum_{i=1}^N (\frac{1}{N})^2} \sum_{i=1}^N (x_i - \bar{x})^2 \frac{1}{N}} &= \sqrt{\frac{1}{1 - \frac{1}{N}} \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} \\
&= \sqrt{\frac{1}{\frac{N-1}{N}} \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} \\
&= \sqrt{\frac{N}{N-1} \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} \\
&= \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}
\end{aligned}$$

We see that we end up with the usual “sample” standard deviation.

A.1 Bias of Arbitrary Sample Moments.

We now proceed in the same way as above to compute the bias of more general moments. Throughout this subsection we assume the central moments $M_k \equiv \mathbb{E}[(x_i - \mu)^k]$ exist (are finite) for all $k \leq r$. The strategy mirrors the second-moment derivation: rewrite $x_i - \bar{x} = (x_i - \mu) + (\mu - \bar{x})$ and expand the r^{th} power via the binomial theorem. The resulting power of $\sum_l (x_l - \mu)w_l$ is then expanded by the multinomial theorem, splitting off the $l = i$ term so that independence may be used when the expectation is distributed across the product.

$$\begin{aligned}
\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^r w_i \right] &= \sum_{i=1}^N \mathbb{E} [(x_i - \bar{x})^r] w_i \\
&= \sum_{i=1}^N \mathbb{E} [(x_i - \mu) + (\mu - \bar{x})]^r w_i \\
&= \sum_{i=1}^N \mathbb{E} \left[\sum_{j=0}^r (-1)^j \binom{r}{j} (x_i - \mu)^{r-j} (\bar{x} - \mu)^j \right] w_i \\
&= \sum_{i=1}^N \sum_{j=0}^r (-1)^j \binom{r}{j} \mathbb{E} [(x_i - \mu)^{r-j} (\bar{x} - \mu)^j] w_i \\
&= \sum_{i=1}^N \sum_{j=0}^r (-1)^j \binom{r}{j} \mathbb{E} \left[(x_i - \mu)^{r-j} \left(\sum_{l=1}^N (x_l - \mu) w_l \right)^j \right] w_i \\
&= \sum_{i=1}^N \sum_{j=0}^r (-1)^j \binom{r}{j} \mathbb{E} \left[(x_i - \mu)^{r-j} \left((x_i - \mu) w_i + \sum_{\substack{l \in [1, N] \\ l \neq i}} (x_l - \mu) w_l \right)^j \right] w_i \\
&= \sum_{i=1}^N \sum_{j=0}^r (-1)^j \binom{r}{j} \mathbb{E} \left[(x_i - \mu)^{r-j} \sum_{k=0}^j \binom{j}{k} (x_i - \mu)^{j-k} w_i^{j-k} \left(\sum_{\substack{l \in [1, N] \\ l \neq i}} (x_l - \mu) w_l \right)^k \right] w_i \\
&= \sum_{i=1}^N \sum_{j=0}^r \sum_{k=0}^j (-1)^j \binom{r}{j} \binom{j}{k} \mathbb{E} \left[(x_i - \mu)^{r-k} w_i^{j-k} \left(\sum_{\substack{l \in [1, N] \\ l \neq i}} (x_l - \mu) w_l \right)^k \right] w_i \\
&= \sum_{i=1}^N \sum_{j=0}^r \sum_{k=0}^j (-1)^j \binom{r}{j} \binom{j}{k} \mathbb{E} [(x_i - \mu)^{r-k}] \mathbb{E} \left[\left(\sum_{\substack{l \in [1, N] \\ l \neq i}} (x_l - \mu) w_l \right)^k \right] w_i^{j+1-k} \\
&= \sum_{i=1}^N \sum_{j=0}^r (-1)^j \binom{r}{j} \mathbb{E} [(x_i - \mu)^r] w_i^{j+1} \\
&\quad + \sum_{i=1}^N \sum_{j=2}^r \sum_{k=2}^j (-1)^j \binom{r}{j} \binom{j}{k} \mathbb{E} [(x_i - \mu)^{r-k}] \mathbb{E} \left[\left(\sum_{\substack{l \in [1, N] \\ l \neq i}} (x_l - \mu) w_l \right)^k \right] w_i^{j+1-k}
\end{aligned} \tag{41}$$

Separating out the $j = 0$ term (which yields the bare M_r) and noting the $k = 1$ contributions vanish because

$\mathbb{E}[x_l - \mu] = 0$, we restrict the inner multinomial sum to indices with $k_n \neq 1$ and obtain

$$\begin{aligned}
\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^r w_i \right] &= M_r + M_r \sum_{j=1}^r (-1)^j \binom{r}{j} \sum_{i=1}^N w_i^{j+1} + \sum_{i=1}^N \sum_{j=2}^r \sum_{k=2}^j (-1)^j \binom{r}{j} \binom{j}{k} M_{r-k} \\
&\quad \times \mathbb{E} \left[\sum_{\substack{k_1+k_2+\dots+k_N=k \\ \forall n, k_n \geq 0 \\ k_i=0}} \binom{k}{k_1, k_2, \dots, k_N} \prod_{\substack{l \in [1, N] \\ l \neq i}} (x_l - \mu)^{k_l} w_l^{k_l} \right] w_i^{j+1-k} \\
&= M_r + M_r \sum_{j=1}^r (-1)^j \binom{r}{j} \sum_{i=1}^N w_i^{j+1} + \sum_{i=1}^N \sum_{j=2}^r \sum_{k=2}^j (-1)^j \binom{r}{j} \binom{j}{k} M_{r-k} \\
&\quad \times \sum_{\substack{k_1+k_2+\dots+k_N=k \\ \forall n, k_n \geq 0 \wedge k_n \neq 1 \\ k_i=0}} \binom{k}{k_1, k_2, \dots, k_N} \left(\prod_{\substack{l \in [1, N] \\ l \neq i}} \mathbb{E}[(x_l - \mu)^{k_l}] w_l^{k_l} \right) w_i^{j+1-k} \\
&= M_r + M_r \sum_{j=1}^r (-1)^j \binom{r}{j} \sum_{i=1}^N w_i^{j+1} + \sum_{i=1}^N \sum_{j=2}^r \sum_{k=2}^j (-1)^j \binom{r}{j} \binom{j}{k} M_{r-k} \\
&\quad \times \sum_{\substack{k_1+k_2+\dots+k_N=k \\ \forall n, k_n \geq 0 \wedge k_n \neq 1 \\ k_i=0}} \binom{k}{k_1, k_2, \dots, k_N} \left(\prod_{l=1}^N M_{k_l} w_l^{k_l} \right) w_i^{j+1-k}
\end{aligned} \tag{42}$$

Here, we use the notation $M_r \equiv \mathbb{E}[(x_i - \mu)^r]$ for the r^{th} central moment of the distribution of the $\{x_i\}_{i=1}^N$ (so that $M_2 = \sigma^2$).

We now drop the $k_i = 0$ constraint by allowing k_i to take any value in $\{0\} \cup \{2, 3, \dots, k\}$ – this makes the *first* inner sum below independent of i – and subtract back the $k_i \in \{2, \dots, k\}$ contributions as a separate z -indexed sum, with $z = k_i$. In the subtracted term, the remaining $N - 1$ slots $k_1, \dots, k_{N-1}$ correspond to the indices $l \neq i$, relabeled as $1, \dots, N - 1$; that term therefore still depends on i , both through this exclusion and through the factor $w_i^{j+z+1-k}$. The multinomial coefficient with one slot equal to z factors as

$\binom{k}{z} \binom{k-z}{k_1, \dots, k_{N-1}}$, which produces the $\binom{k}{z}$ prefactor in the subtraction below.¹⁶

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^r w_i \right] &= M_r + M_r \sum_{j=1}^r (-1)^j \binom{r}{j} \sum_{i=1}^N w_i^{j+1} + \sum_{j=2}^r \sum_{k=2}^j (-1)^j \binom{r}{j} \binom{j}{k} M_{r-k} \\ &\quad \times \left[\sum_{i=1}^N w_i^{j+1-k} \left(\sum_{\substack{k_1+k_2+\dots+k_N=k \\ \forall n, k_n \geq 0 \wedge k_n \neq 1}} \binom{k}{k_1, k_2, \dots, k_N} \prod_{l=1}^N M_{k_l} w_l^{k_l} \right) \right. \\ &\quad \left. - \sum_{z=2}^k \binom{k}{z} M_z \left(\sum_{i=1}^N w_i^{j+z+1-k} \sum_{\substack{k_1+k_2+\dots+k_{N-1}=k-z \\ \forall n, k_n \geq 0 \wedge k_n \neq 1}} \binom{k-z}{k_1, k_2, \dots, k_{N-1}} \prod_{l=1}^{N-1} M_{k_l} w_l^{k_l} \right) \right] \end{aligned} \quad (43)$$

We check the case $r = 2$ to see if it coincides with the previous section:

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^2 w_i \right] &= \sigma^2 + \sigma^2 \sum_{j=1}^2 (-1)^j \binom{2}{j} \sum_{i=1}^N w_i^{j+1} + \sum_{j=2}^2 \sum_{k=2}^j (-1)^j \binom{2}{j} \binom{j}{k} M_{2-k} \\ &\quad \times \sum_{i=1}^N \left[\sum_{\substack{k_1+k_2+\dots+k_N=k \\ \forall n, k_n \geq 0 \wedge k_n \neq 1}} \binom{k}{k_1, k_2, \dots, k_N} \left(\prod_{l=1}^N M_{k_l} w_l^{k_l} \right) w_i^{j+1-k} \right. \\ &\quad \left. - \sum_{z=2}^k \binom{k}{z} M_z \sum_{\substack{k_1+k_2+\dots+k_{N-1}=k-z \\ \forall n, k_n \geq 0 \wedge k_n \neq 1}} \binom{k-z}{k_1, k_2, \dots, k_{N-1}} \left(\prod_{l=1}^{N-1} M_{k_l} w_l^{k_l} \right) w_i^{j+z+1-k} \right] \end{aligned} \quad (44)$$

This becomes

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^2 w_i \right] &= \sigma^2 + \sigma^2 \sum_{j=1}^2 (-1)^j \binom{2}{j} \sum_{i=1}^N w_i^{j+1} \\ &\quad + \sum_{\substack{k_1+k_2+\dots+k_N=2 \\ \forall n, k_n \geq 0 \wedge k_n \neq 1}} \binom{2}{k_1, k_2, \dots, k_N} \left(\prod_{l=1}^N M_{k_l} w_l^{k_l} \right) \left(\sum_{i=1}^N w_i \right) - \sigma^2 \left(\sum_{i=1}^N w_i^3 \right) \end{aligned} \quad (45)$$

Which is

$$\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^2 w_i \right] = \sigma^2 + \sigma^2 \left(-2 \sum_{i=1}^N w_i^2 + \sum_{i=1}^N w_i^3 \right) + \sigma^2 \sum_{l=1}^N w_l^2 \left(\sum_{i=1}^N w_i \right) - \sigma^2 \left(\sum_{i=1}^N w_i^3 \right) \quad (46)$$

¹⁶We use the standard interpretation of any sum of the form, $\sum_{i=n}^m f(i)$, to be 0 if $m < n$.

This becomes

$$\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^2 w_i \right] = \sigma^2 \left(1 - \sum_{i=1}^N w_i^2 \right) \quad (47)$$

As a more complicated example we try $r = 3$. In this case the formula is:

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^3 w_i \right] &= M_3 + M_3 \sum_{j=1}^3 (-1)^j \binom{3}{j} \sum_{i=1}^N w_i^{j+1} + \sum_{j=2}^3 \sum_{k=2}^j (-1)^j \binom{3}{j} \binom{j}{k} M_{3-k} \\ &\quad \times \sum_{i=1}^N \left[\sum_{\substack{k_1+k_2+\dots+k_N=k \\ \forall n, k_n \geq 0 \wedge k_n \neq 1}} \binom{k}{k_1, k_2, \dots, k_N} \left(\prod_{l=1}^N M_{k_l} w_l^{k_l} \right) w_i^{j+1-k} \right. \\ &\quad \left. - \sum_{z=2}^k \binom{k}{z} M_z \sum_{\substack{k_1+k_2+\dots+k_{N-1}=k-z \\ \forall n, k_n \geq 0 \wedge k_n \neq 1}} \binom{k-z}{k_1, k_2, \dots, k_{N-1}} \left(\prod_{l=1}^{N-1} M_{k_l} w_l^{k_l} \right) w_i^{j+z+1-k} \right] \end{aligned} \quad (48)$$

Which is

$$\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^3 w_i \right] = M_3 + M_3 \left(-3 \sum_{i=1}^N w_i^2 + 3 \sum_{i=1}^N w_i^3 - \sum_{i=1}^N w_i^4 \right) - M_3 \left(\sum_{l=1}^N w_l^3 \left(\sum_{i=1}^N w_i \right) - \sum_{i=1}^N w_i^4 \right) \quad (49)$$

Or,

$$\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^3 w_i \right] = M_3 \left(1 - 3 \sum_{i=1}^N w_i^2 + 2 \sum_{i=1}^N w_i^3 \right) \quad (50)$$

Therefore, the corresponding unbiased estimator for the 3rd central moment is:¹⁷

$$\frac{\sum_{i=1}^N (x_i - \bar{x})^3 w_i}{1 - 3 \sum_{i=1}^N w_i^2 + 2 \sum_{i=1}^N w_i^3} \quad (51)$$

For $r > 3$ the bookkeeping is more involved. We illustrate the case $r = 4$ below. Substituting directly into the general formula above gets unwieldy; we instead organize the computation by the value of k , the power of $\sum_{l \neq i} (x_l - \mu) w_l$ that appears.

Let $y_i = x_i - \mu$ and $z = \bar{x} - \mu = \sum_{l=1}^N y_l w_l$. Split $z = y_i w_i + s_i$, where

$$s_i \equiv \sum_{\substack{l \in [1, N] \\ l \neq i}} y_l w_l$$

¹⁷As mentioned in the case of the second moment, this formula does not apply when the weights are concentrated at a single index. Consequently, we must have $N > 1$.

is independent of y_i . Expanding $(x_i - \bar{x})^4 = (y_i - z)^4$ via the binomial theorem, and then $z^k = (y_i w_i + s_i)^k$ via a second application, gives:

$$\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^4 w_i \right] = \sum_{j=0}^4 \sum_{k=0}^j (-1)^j \binom{4}{j} \binom{j}{k} M_{4-k} \sum_{i=1}^N w_i^{j-k+1} \mathbb{E} [s_i^k]. \quad (52)$$

Let $W_j = \sum_{i=1}^N w_i^j$. We collect the contributions by the value of k :

- $k = 0$: $\mathbb{E} [s_i^0] = 1$, giving $M_4 \sum_{j=0}^4 (-1)^j \binom{4}{j} W_{j+1} = M_4(1 - 4W_2 + 6W_3 - 4W_4 + W_5)$.
- $k = 1$: $\mathbb{E} [s_i^1] = 0$.
- $k = 2$: $\mathbb{E} [s_i^2] = M_2(W_2 - w_i^2)$, $M_{4-k} = M_2$, summed over $j \in \{2, 3, 4\}$: $M_2^2 [6(W_2 - W_3) - 12(W_2^2 - W_4) + 6(W_2 W_3 - W_5)]$.
- $k = 3$: $M_{4-k} = M_1 = 0$.
- $k = 4$: only $j = 4$ contributes, with $\mathbb{E} [s_i^4] = M_4(W_4 - w_i^4) + 6M_2^2 [\frac{1}{2}(W_2^2 - W_4) - w_i^2 W_2 + w_i^4]$, giving $M_4(W_4 - W_5) + 6M_2^2 [\frac{1}{2}(W_2^2 - W_4) - W_2 W_3 + W_5]$.

Adding these (the W_5 terms cancel in the M_4 piece; the $W_2 W_3$ and W_5 terms cancel in the M_2^2 piece, and $6 \cdot \frac{1}{2}(W_2^2 - W_4) = 3W_2^2 - 3W_4$):

$$\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^4 w_i \right] = M_4(1 - 4W_2 + 6W_3 - 3W_4) + 3M_2^2(2W_2 - 2W_3 - 3W_2^2 + 3W_4). \quad (53)$$

As a check, at $N = 2$ with $w_1 = w_2 = \frac{1}{2}$ ($W_2 = \frac{1}{2}$, $W_3 = \frac{1}{4}$, $W_4 = \frac{1}{8}$): the M_4 coefficient is $1 - 2 + \frac{3}{2} - \frac{3}{8} = \frac{1}{8}$, and the M_2^2 coefficient is $3(1 - \frac{1}{2} - \frac{3}{4} + \frac{3}{8}) = \frac{3}{8}$, matching the direct evaluation $\frac{M_4}{8} + \frac{3M_2^2}{8}$.

The corresponding relation for the kurtosis, M_4/M_2^2 , is:

$$\frac{M_4}{M_2^2} = \frac{\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^4 w_i \right] / M_2^2 - 3(2W_2 - 2W_3 - 3W_2^2 + 3W_4)}{1 - 4W_2 + 6W_3 - 3W_4}. \quad (54)$$

Since we don't know $M_2 = \sigma^2$, we use our unbiased estimate $\widehat{M}_2 = \frac{\sum_{j=1}^N (x_j - \bar{x})^2 w_j}{1 - W_2}$. We also replace $\mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^4 w_i \right]$ with the sample estimate $\sum_{i=1}^N (x_i - \bar{x})^4 w_i$. Our estimate of the kurtosis is then:

$$\frac{M_4}{M_2^2} \approx \frac{\frac{(1 - W_2)^2 \sum_{i=1}^N (x_i - \bar{x})^4 w_i}{\left(\sum_{j=1}^N (x_j - \bar{x})^2 w_j \right)^2} - 3(2W_2 - 2W_3 - 3W_2^2 + 3W_4)}{1 - 4W_2 + 6W_3 - 3W_4}. \quad (55)$$